Examiner only

Answer **all** *questions in the spaces provided.*

1. Drinks manufacturers use standard food colourings which are identified by E-numbers. A student was asked to investigate which E-numbers were present in different brands of orange drinks.

The student suggested the following method:

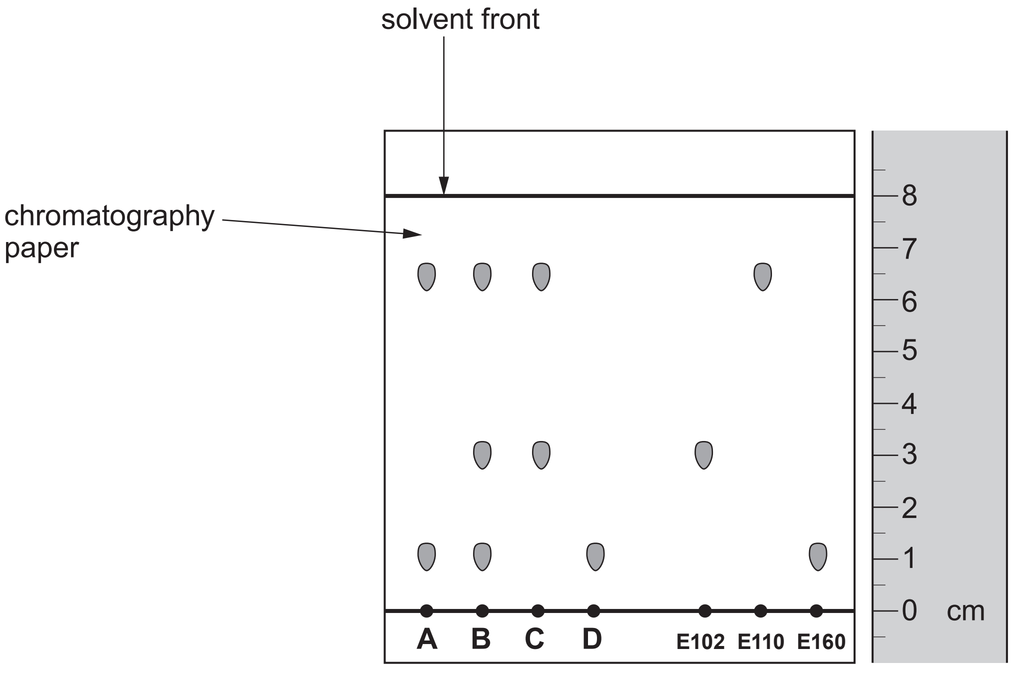
1. Place spots of each drink **A**, **B**, **C** and **D** on a pen line on chromatography paper.
2. Place spots of three common food colourings, **E102**, **E110**, **E160** on the pen line.
3. Hang the chromatography paper in a beaker and add water until it is above the pen line.
4. Leave the paper to stand in water in a beaker until the water rises to near the top of the chromatography paper.

(a) The student failed to produce a chromatogram using this method. State **two** errors in this method. [2]

1. ....

2. ....

(b) Another student, using a valid method, obtained the chromatogram below *(not drawn to scale)*.



Answer the following questions using the information in the chromatogram.

(i) State which food colouring (**E102**, **E110**, **E160**) is the most soluble. [1]

....

(ii) State which drinks (**A**, **B**, **C** or **D**) contained **E110**. [1]

....

Examiner only

(iii) The student predicted that all orange drinks contain the same E-numbers. Explain whether this prediction was correct by comparing the food colourings found in drinks **B** and **D**. [2]

....

....

....

(iv) The retention factor (R_f) value of a substance can be used to identify that substance from a chromatogram.

I. State the distance travelled by the solvent front. [1]

Distance = ... cm

II. Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by the solvent front}}$$

to calculate the R_f value for **E102**. [2]

R_f = ...

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9